MGSC 416- DATA-DRIVEN MODELS FOR OPERATIONS ANALYTICS

PROJECT GUIDELINES

Presentation Deck Due April 2nd, 11:59pm on EdStem/MyCourses

Report + code due April 2nd, 11:59pm on EdStem/MyCourses

OBJECTIVE

The goal behind the project in this course is to provide an experiential learning component to how data driven methods (predictive, prescriptive, and descriptive) can be applied to real world problems with available, sometimes messy data.

TOPIC/DATASET

You are free to choose any topic of your preference, whatever you are passionate about. I have listed at the end of this document some links to potential datasets, as well as a list of some of the data-driven projects ideas of past students, but of course you can be creative and look for the dataset/problem that interests you.

My only advice here is to be mindful of what dataset is available to you and how the choice/limitation of dataset that you choose can dictate the analytics (optimization, prediction etc....) directions you can take in this project. Both creativity and the ability to assess the weakness and the strength of the dataset with respect to the chosen analytical problem are parts of how the project will be assessed

MODEL/TECHNIQUES

The project should rely on the techniques we learned the course. This means that your project should incorporate each of these components:

1. Prediction Techniques: (Examples: Regression Analysis, Time Series)
2. Optimization techniques: (Examples: Linear programming, Non-linear Programming, Integer Programming)

We have also learned interesting ideas from the case studies and business contexts we have seen in the course, for example on how to apply RM concepts (The use of Bid Prices or Shadow Prices, Capacity/Budget Allocations Problems, Pricing Problems), on how to correct autocorrelation errors in regression (dynamic regression), tricks in incorporating business constraints and managerial insights in optimization problems (Fantasy Sports, Kidney Exchange...). You're welcome to take some of the ideas and incorporate them in your analysis if it helps advance the analysis.

You may also want to incorporate Descriptive analysis (Examples: summary statistics, Data imputation, Data Visualization, Network Analysis) in your project. A proper use of such methods can sometimes guide your choice of problem and subsequent analysis.

You can also use, of course, complementary prediction or prescriptive techniques from other classes you have taken in Business Analytics, but I expect your analysis to incorporate some concepts of the course. I also expect you to demonstrate that you have mastered the proper use of these topics for new problems and datasets.
Overall, you will be assessed on the proper use of any techniques you use on your dataset for the chosen problem.

DATASET REPOSITORIES

·You can choose any dataset you would like. You may want to spend an afternoon searching for something interesting and appropriate. There are also some sites (like Kaggle: https://www.kaggle.com) where you can find many interesting datasets

DELIVERABLES

You are required to work in teams (4-5 students per team). The end goal is to provide the followings:

1. A Pitch deck (of up to 6 slides (not counting first introductory slide)), where you would (i) pitch the problem, (ii) present an overview of the data, and (iii) the idea behind the analysis/solution. You will be assessed on completion of this assignment and the presence of these three components on your slides. On presentation day, all teams will present their slides in front of the class, and we collectively will vote for their 3-4 favorite presentations/pitches. The winners will get a special prize.

2. A summary note (report of 6 pages maximum) accompanied with appendices (up to 10 pages) which can include tables, graphs, some code outputs etc... The report should be typed and written in a clear and concise manner. The report should include the effectiveness of the problem formulation, the ensuing analysis, and the final recommendation. You should also submit the related code to your analysis (In Python).

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HOW TO SHAPE YOUR REPORT?

To organize your report, you are welcome to adopt the following framework, although at the end you are free to deviate if you deem it necessary:

INTRODUCTION (1/2-1 PAGE)

Explain and motivate the project's problem and goals.

DATA OVERVIEW (UP TO 1 PAGE)

Describe the dataset, which features, size, any interesting changes/data wrangling you have done (for example merging datasets, creating new features....), showcase any interesting summary statistics that might guide the choice of your methodology.

METHODOLOGY/MODEL (1 AND 1/2 PAGES)

Explain what methodologies/model analysis you are using, and why they are suitable for the problem.

RESULTS AND INSIGHTS (1-2 PAGE)

Present your findings.

CONCLUSION AND LIMITATIONS (1 PAGE)

Summarize the results, discuss any business/managerial insights that it provides, discuss critically the limitations of the analysis/data with respect to the findings.

APPENDICES

Should contain all tables and figures, none of which should be in the report main document.

CODE

Please attach at the end an organized PDF document of your code (commented). I strongly suggest using Jupyter Notebook to organize your Python code.

You can use Microsoft Word or Latex or any text editor to organize your report, as long as it is clearly presented and aesthetically pleasing.